

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

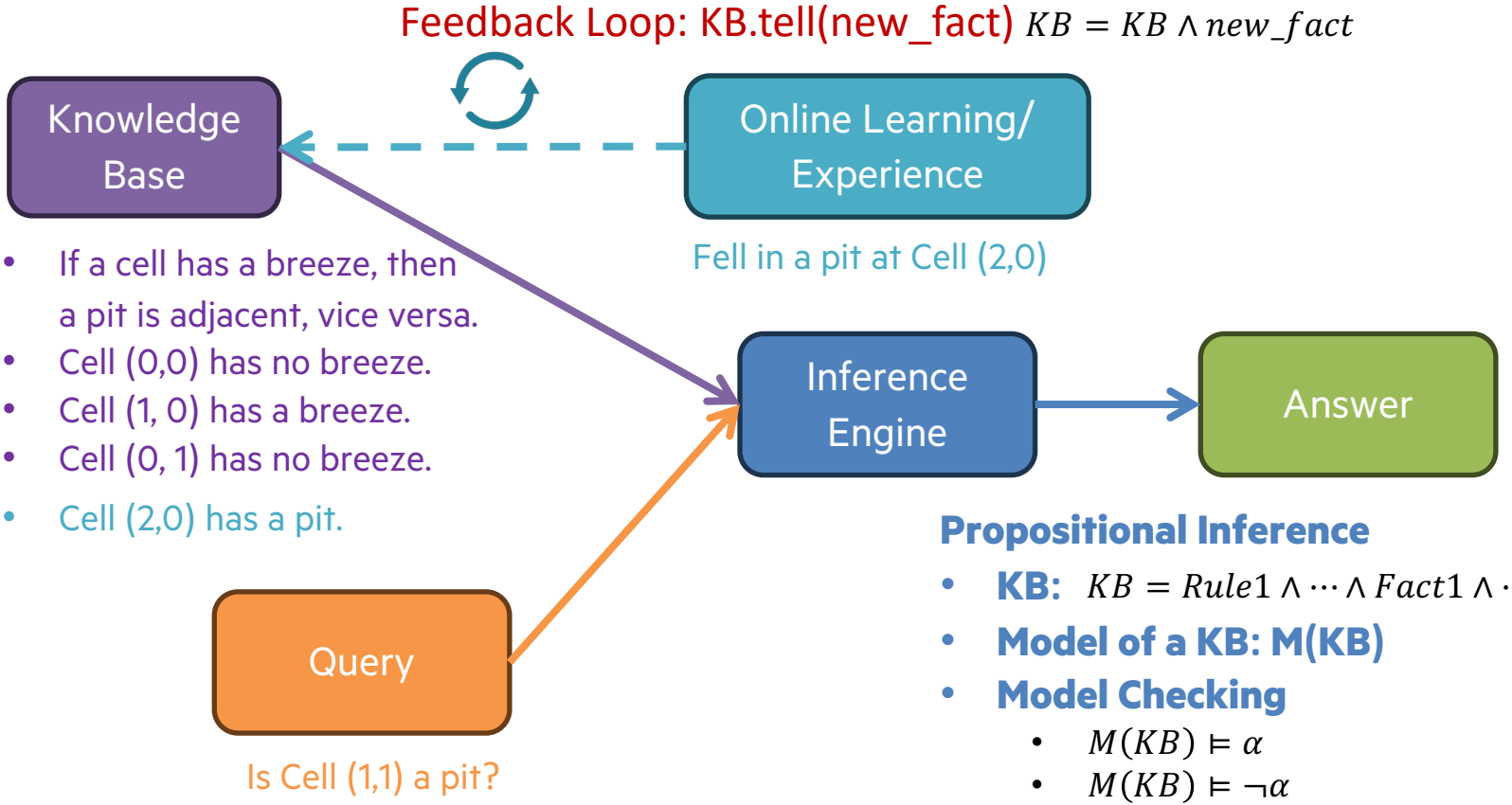
Announcements

- Recitation recordings will be posted on Brightspace (as soon as they are available)
- Classwork 1
  - Due **this Friday, 9/6 @5 pm**
- Classwork 2
  - Available on Brightspace/GitHub Classroom (group-based)
  - Due **next Friday, 9/13 @5 pm**
- Homework 1 will be released next Tuesday

This Lecture

- Theorem Proving
- Knowledge Base Properties
- Resolution
- Classwork 2

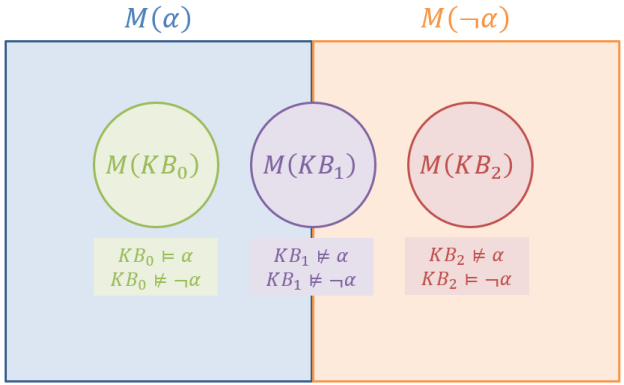
Last Lecture



Model Checking

- Why do we need model checking?
  - Determine if we have **enough information in KB** to make a conclusion
- How to check a model?
  - Entailment ⊨: $\mathrm{KB} \models \alpha$ and $\mathrm{KB} \models \neg\alpha$
  - For a query α,

$$KB \models \alpha \Leftrightarrow M(KB) \subseteq M(\alpha)$$



One Approach – Model Enumeration

World	R	P	S	$KB = (\neg P \vee S) \wedge \neg R$	$\neg R$	$\neg P$
w_0	F	F	F	T	T	T
w_1	F	F	T	T	T	T
w_2	F	T	F	F	T	F
w_3	F	T	T	T	T	F
w_4	T	F	F	F	F	T
w_5	T	F	T	F	F	T
w_6	T	T	F	F	F	F
w_7	T	T	T	F	F	F

What is the limitation of this approach?

$KB \models R?$	No
$KB \models \neg R?$	Yes
$KB \models P?$	No
$KB \models \neg P?$	No

Propositional Theorem Proving

- What
  - The process of showing a KB entails a sentence **without needing** to consult the ~~truth table~~ of the system.
- How
  - Perform inference at the symbolic level

Example

- Two propositions
  - R: whether it is **raining** outside
  - W: whether the sidewalk is **wet**
- KB

1. $R \Rightarrow W$
2. R

---

3. W



(Symbolic)
Inference

Rule 1: Modus Ponens

- A rule of logic asserting that when an **implication's premise** (i.e., P in $P \Rightarrow Q$) is met, then the conclusion (i.e., Q in $P \Rightarrow Q$) may be **inferred** (i.e., added as an inferred fact to the KB).
- Formally,

$$\frac{(\alpha \Rightarrow \beta) \wedge \alpha}{\beta}$$

What we have in a KB

What we we can infer

"if -then"

Example

- Two propositions
  - R: whether it is **raining** outside
  - W: whether the sidewalk is **wet**
- KB

1. $R \Rightarrow W$
2. $\neg W$

---

3. $\neg R$

What can we infer?

Can this be deduced only using MP?

No. We need some rules that are more general.

Knowledge Base Properties

What we know:

$P \Rightarrow Q \equiv \neg P \vee Q$

$$\frac{(\alpha \Rightarrow \beta) \wedge \alpha}{\beta}$$

What are given:

$R \Rightarrow W$

$\neg W$

What we want to prove:

$\neg R$

How?

$$R \Rightarrow W \equiv \neg R \vee W \quad \text{Logical Equivalence}$$

$$\equiv W \vee \neg R \quad \text{Commutativity}$$

$$\equiv \neg W \Rightarrow \neg R \quad \text{Logical Equivalence}$$

$$\frac{\neg W}{\neg R} \quad \text{MP}$$

Conjunctive Normal Form (CNF)

- CNF: A logical sentence consists solely of a **conjunction** of **clauses** with negations on propositions alone.
  - Conjunction: logical AND ($\wedge$)
  - Clause: sentence consists exclusively of disjoined (logical OR ($\vee$)) propositions ($Clause = P_1 \vee P_2 \vee \cdots \vee P_n$)
- Formally,

$$KB = Clause_1 \wedge Clause_2 \wedge \cdots \wedge Clause_k$$

$$KB = (P \vee Q) \wedge (R \vee S \vee \neg T) \wedge (\ldots) \wedge \cdots$$

Conversion to CNF

Any sentences can be converted to CNF in two steps:

- Step 1: **Rephrase** (i.e., $\equiv$) all biconditionals ($\Leftrightarrow$) and implications ($\Rightarrow$) to use only $\neg$, $\vee$, $\wedge$ connectives.
- Step 2: Push all negations in to parentheses so that negation s are only applied to propositions.

Backus-Naur Form (BNF)

$$
\begin{aligned}
\textit{Sentence} &\rightarrow \textit{AtomicSentence} \mid \textit{ComplexSentence} \\
\textit{AtomicSentence} &\rightarrow \textit{True} \mid \textit{False} \mid P \mid Q \mid R \mid \ldots \\
\textit{ComplexSentence} &\rightarrow (\ \textit{Sentence}\ ) \mid [\ \textit{Sentence}\ ] \\
&\mid \neg\ \textit{Sentence} \\
&\mid \textit{Sentence} \wedge \textit{Sentence} \\
&\mid \textit{Sentence} \vee \textit{Sentence} \\
&\mid \textit{Sentence} \Rightarrow \textit{Sentence} \\
&\mid \textit{Sentence} \Leftrightarrow \textit{Sentence} \\
\text{OPERATOR PRECEDENCE} &: \neg, \wedge, \vee, \Rightarrow, \Leftrightarrow
\end{aligned}
$$

Figure 7.7 A BNF (Backus–Naur Form) grammar of sentences in propositional logic, along with operator precedences, from highest to lowest.

Logical Equivalence Rules

$$
\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{De Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{De Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}
$$

Figure 7.11 Standard logical equivalences. The symbols α, β, and γ stand for arbitrary sentences of propositional logic.

Example

Convert the following sentence into CNF:

$$B_{1,0} \Leftrightarrow (P_{2,0} \vee P_{1,1})$$

$$\left(B_{1,0} \Rightarrow (P_{2,0} \vee P_{1,1})\right) \wedge \left((P_{2,0} \vee P_{1,1}) \Rightarrow \mathrm{B}_{1,0}\right)$$

$$\left(\neg B_{1,0} \vee (P_{2,0} \vee P_{1,1})\right) \wedge \left(\neg(P_{2,0} \vee P_{1,1}) \vee \mathrm{B}_{1,0}\right)$$

$$\left(\neg B_{1,0} \vee (P_{2,0} \vee P_{1,1})\right) \wedge \left((\neg P_{2,0} \wedge \neg P_{1,1}) \vee \mathrm{B}_{1,0}\right)$$

$$\left(\neg B_{1,0} \vee (P_{2,0} \vee P_{1,1})\right) \wedge (\neg P_{2,0} \vee \mathrm{B}_{1,0}) \wedge (\neg P_{1,1} \vee \mathrm{B}_{1,0})$$

$$A \Leftrightarrow B \equiv (A \Rightarrow B) \wedge (B \Rightarrow A)$$

$$P \Rightarrow Q \equiv \neg P \vee Q$$

$$\neg(A \vee B) \equiv (\neg A \wedge \neg B)$$

$$(A \wedge B) \vee C \equiv (A \vee C) \wedge (B \vee C)$$

Why do we want CNF?

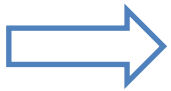
Resolution

- Example:

1. $R \Rightarrow W$
2. $\neg W$

---

3. $\neg R$



1. $\neg R \vee W$
2. $\neg\neg W$

---

3. $\neg R$

Rule 2: Resolution

- **Resolution** is a general form of Modus Ponens that operates on **two clauses**; if the clauses **disagree on a proposition** P_i (i.e., one has P_i and the other has its negation $\neg P_i$), then **a new clause** can be inferred consisting of the **union of propositions** between the clauses **except the disagreed-upon.**
- Formally,

$$\frac{(P_1 \vee P_2 \vee P_i \vee \cdots) \wedge (P_3 \vee P_4 \vee \neg P_i \vee \cdots)}{(P_1 \vee P_2 \vee P_3 \vee P_4 \vee \cdots)}$$

Example

$Clause_1$	$Clause_2$	$Clause_1 \wedge Clause_2$
$\neg R \vee S$	R	S
$S \vee \neg R \vee T$	$Q \vee R$	$S \vee T \vee Q$
$\neg S$	S	$\emptyset$ Contradiction
$\neg S \vee R$	$S \vee \neg R$	$S \vee \neg S \equiv R \vee \neg R \equiv True$ Not a Contradiction

Resolution: an operation defined on a *single* literal at a time!

CNF and Resolution

- CNF $KB = Clause_1 \wedge Clause_2 \wedge \cdots \wedge Clause_k$

  $KB = (P \vee Q) \wedge (R \vee S \vee \neg T) \wedge (\dots) \wedge \cdots$

- Resolution

$$\frac{(P_1 \vee P_2 \vee P_i \vee \cdots) \wedge (P_3 \vee P_4 \vee \neg P_i \vee \cdots)}{(P_1 \vee P_2 \vee P_3 \vee P_4 \vee \cdots)}$$

- How to use CNF and Resolution in an inference procedure?

Proof by Contradiction

1. Assume to the contrary (i.e., assume α is true)

$$KB = KB \wedge \neg\alpha$$

2. Try to derive a contradiction
   - Apply pairwise resolution to all clauses, including those that are inferred by resolution until:
     - $KB \wedge \neg\alpha \equiv \emptyset$ (reach a contradiction). Therefore, $KB \models \alpha$.
     - There are no other resolutions to consider (do not reach a contradiction). Therefore, $KB \not\models \alpha$.

Example

- Query: Is cell (1, 0) safe? ($\neg P_{1,0}$?)
- KB
  1. $\neg B_{0,0} \vee P_{1,0} \vee P_{0,1}$
  2. $\neg P_{1,0} \vee B_{0,0}$
  3. $\neg P_{0,1} \vee B_{0,0}$
  4. $\neg B_{0,0}$

Example

1. $\neg B_{0,0} \vee P_{1,0} \vee P_{0,1}$
2. $\neg P_{1,0} \vee B_{0,0}$
3. $\neg P_{0,1} \vee B_{0,0}$
4. $\neg B_{0,0}$

Step 1: Assume:

5. $P_{1,0}$

Step 2: Iteratively apply resolution until we either reach a contradiction or run out of resolutions to try:

6. $B_{0,0}$ (from clause 1 and 5)
7. $\emptyset$ (from clause 4 and 6)

Therefore, $KB \models \neg P_{0,0}$

Remaining of the lecture

CLASSWORK 2